

Lecture Slide – 15

Euler's Method, Heun's Method &
Midpoint Method

Motivation: Why Do We Need Numerical ODE Solvers?

The General Problem

We seek to solve the Initial Value Problem (IVP):

$$dy/dx = f(x, y), \quad y(x_0) = y_0$$

where $f(x, y)$ can be any function.

Most real-world ODEs have NO closed-form analytical solution. Numerical methods approximate the solution at discrete points.

Real-World Applications



Population dynamics (biology)



Mechanical vibrations (engineering)



Drug concentration in blood



Newton's Law of Cooling



Orbital mechanics (space)



RC/RL electrical circuits

Key Idea: Starting from $y(x_0) = y_0$, we step forward in x to approximate y at $x_1, x_2, \dots$ using the slope $f(x, y)$. The **Runge-Kutta family** achieves high accuracy with fewer function evaluations.

Euler's Method

We can solve ordinary differential equations of the form

$$\frac{dy}{dx} = f(x, y)$$

We can use a numerical method to solve such an equation for the velocity of the falling parachutist. Recall that the method was of the general form

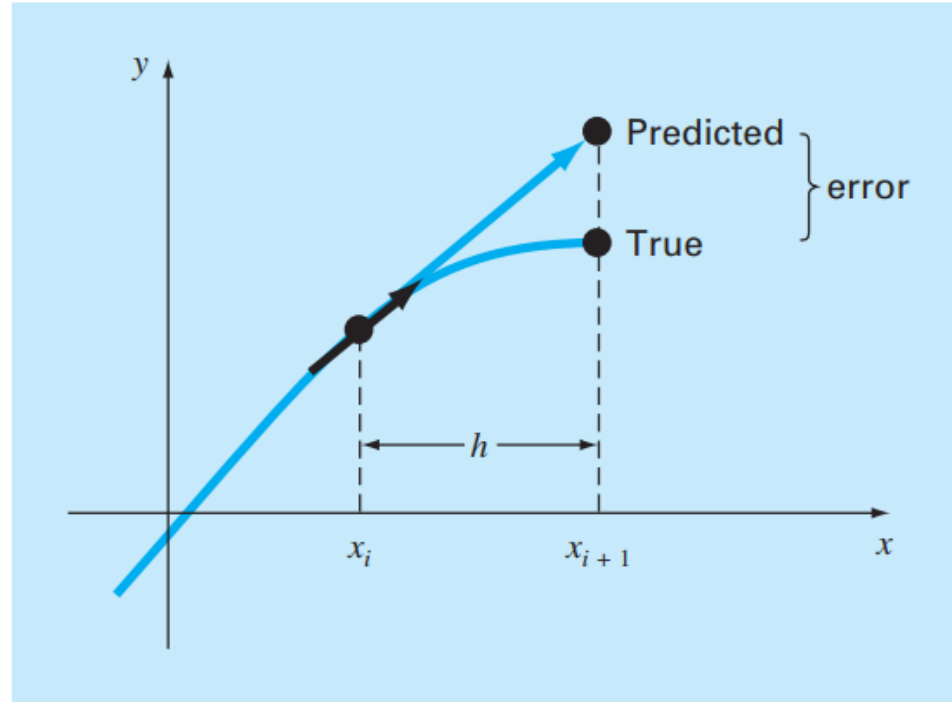
$$\text{New value} = \text{old value} + \text{slope} \times \text{step size}$$

or, in mathematical terms,

$$y_{i+1} = y_i + \phi h$$

According to this equation, the slope estimate of f is used to extrapolate from an old value y_i to a new value y_{i+1} over a distance h .

Euler's Method



Euler's Method

The first derivative provides a direct estimate of the slope at x_i

$$\phi = f(x_i, y_i)$$

where $f(x_i, y_i)$ is the differential equation evaluated at x_i and y_i .

$$y_{i+1} = y_i + f(x_i, y_i)h$$

This formula is referred to as ***Euler's*** (or the *Euler-Cauchy* or the *point-slope*) *method*. A new value of y is predicted using the **slope (equal to the first derivative at the original value of x)** to extrapolate linearly over the step size h .

Example 1

Problem Statement. Use Euler's method to numerically integrate

$$\frac{dy}{dx} = -2x^3 + 12x^2 - 20x + 8.5$$

from $x = 0$ to $x = 4$ with a step size of 0.5. The initial condition at $x = 0$ is $y = 1$.

The exact solution is given by

$$y = -0.5x^4 + 4x^3 - 10x^2 + 8.5x + 1$$

Solution. To implement Euler's method, we can write equation as follows

$$y(0.5) = y(0) + f(0, 1)0.5$$

where $y(0) = 1$ and the slope estimate at $x = 0$ is

Therefore,

$$f(0, 1) = -2(0)^3 + 12(0)^2 - 20(0) + 8.5 = 8.5 \quad y(0.5) = 1.0 + 8.5(0.5) = 5.25$$

Example 1 (cont.)

The true solution at $x = 0.5$ is

$$y = -0.5(0.5)^4 + 4(0.5)^3 - 10(0.5)^2 + 8.5(0.5) + 1 = 3.21875$$

Thus, the error is

$$E_t = \text{true} - \text{approximate} = 3.21875 - 5.25 = -2.03125$$

or, expressed as percent relative error, $et = -63.1\%$. For the second step,

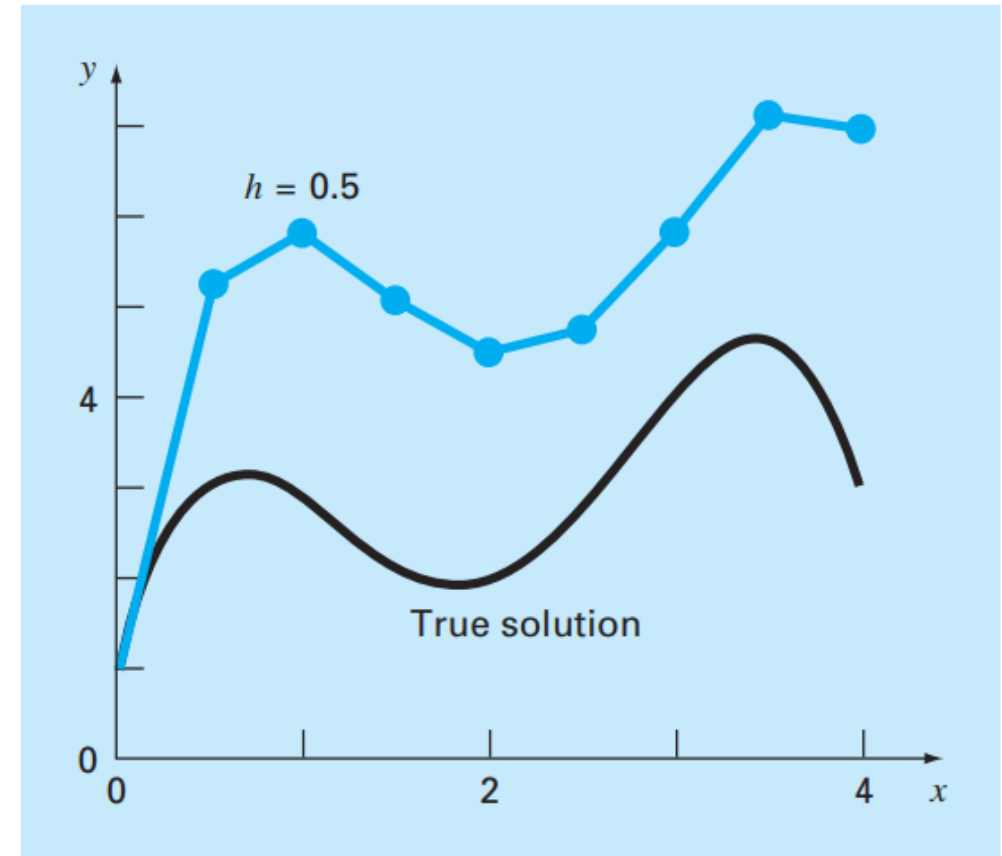
$$\begin{aligned} y(1) &= y(0.5) + f(0.5, 5.25)0.5 \\ &= 5.25 + [-2(0.5)^3 + 12(0.5)^2 - 20(0.5) + 8.5]0.5 \\ &= 5.875 \end{aligned}$$

The true solution at $x = 1.0$ is 3.0, and therefore, the percent relative error is -95.8%.

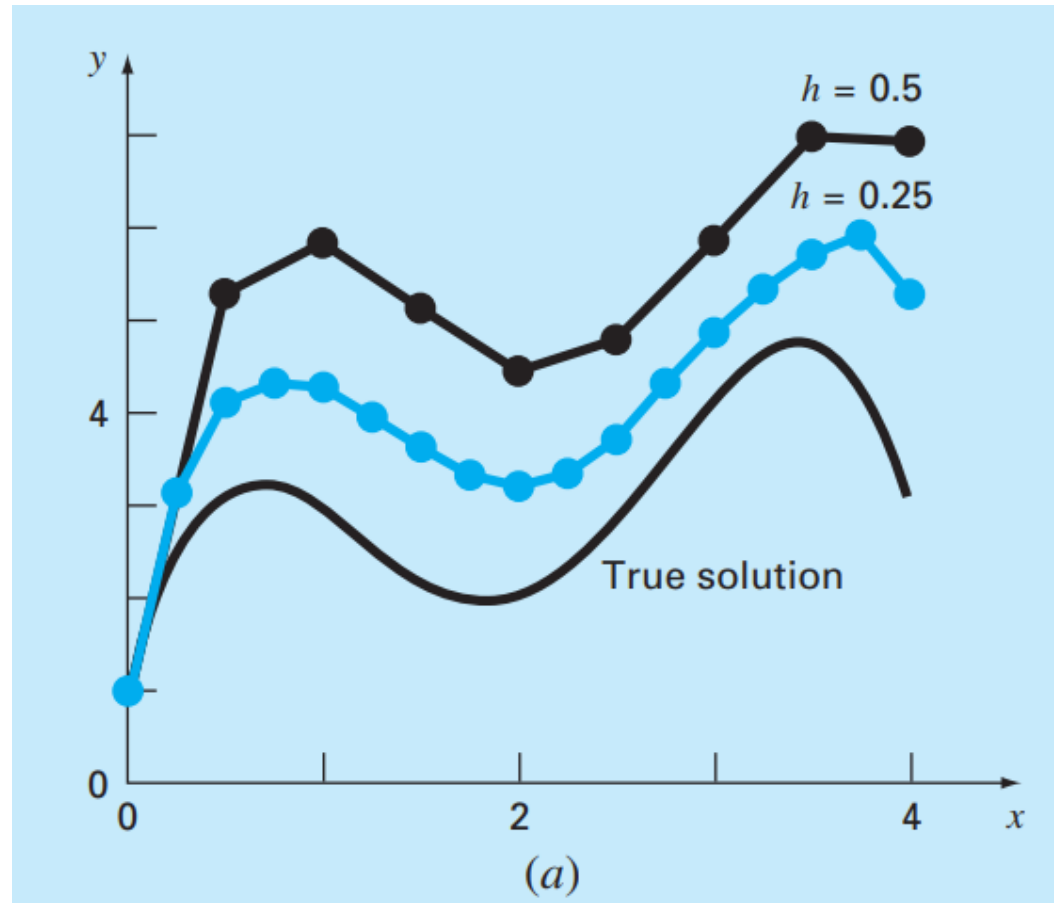
Example 1 (cont.)

Comparison of true and approximate values of the integral of $y' = -2x^3 + 12x^2 - 20x + 8.5$, with the initial condition that $y = 1$ at $x = 0$. The approximate values were computed using Euler's method with a step size of 0.5.

x	y_{true}	y_{Euler}
0.0	1.00000	1.00000
0.5	3.21875	5.25000
1.0	3.00000	5.87500
1.5	2.21875	5.12500
2.0	2.00000	4.50000
2.5	2.71875	4.75000
3.0	4.00000	5.87500
3.5	4.71875	7.12500
4.0	3.00000	7.00000



Example 1 (cont.)



Comparison of two numerical solutions with Euler's method using step sizes of 0.5 and 0.25

Example 2

Consider the **Initial Value Problem (IVP)**

$$y' + 2y = 2 - e^{-4t} \quad y(0) = 1$$

Use **Euler's Method** with a step size of $h=0.1$ to find approximate values of the solution at $t = 0.1, 0.2, 0.3, 0.4,$ and 0.5 . Compare them to the exact values of the solution at these points.

$$y(t) = 1 + \frac{1}{2}e^{-4t} - \frac{1}{2}e^{-2t}$$

In order to use Euler's Method we first need to rewrite the differential equation

$$y' = 2 - e^{-4t} - 2y$$

From this we can see that $f(t, y) = 2 - e^{-4t} - 2y$. Also note that $t_0 = 0$ and $y_0 = 1$.

Example 2 (cont.)

We can now start doing some computations.

$$f_0 = f(0, 1) = 2 - e^{-4(0)} - 2(1) = -1$$

$$y_1 = y_0 + h f_0 = 1 + (0.1)(-1) = 0.9$$

So, the approximation to the solution at $t_1 = 0.1$ is $y_1 = 0.9$.

At the next step we have

$$f_1 = f(0.1, 0.9) = 2 - e^{-4(0.1)} - 2(0.9) = -0.470320046$$

$$y_2 = y_1 + h f_1 = 0.9 + (0.1)(-0.470320046) = 0.852967995$$

Therefore, the approximation to the solution at $t_2 = 0.2$ is $y_2 = 0.852967995$.

Similarly

$$f_2 = -0.155264954$$

$$y_3 = 0.837441500$$

$$f_3 = 0.023922788$$

$$y_4 = 0.839833779$$

$$f_4 = 0.1184359245$$

$$y_5 = 0.851677371$$

Example 2 (cont.)

Time, t_n	Approximation	Exact	Error
$t_0 = 0$	$y_0 = 1$	$y(0) = 1$	0 %
$t_1 = 0.1$	$y_1 = 0.9$	$y(0.1) = 0.925794646$	2.79 %
$t_2 = 0.2$	$y_2 = 0.852967995$	$y(0.2) = 0.889504459$	4.11 %
$t_3 = 0.3$	$y_3 = 0.837441500$	$y(0.3) = 0.876191288$	4.42 %
$t_4 = 0.4$	$y_4 = 0.839833779$	$y(0.4) = 0.876283777$	4.16 %
$t_5 = 0.5$	$y_5 = 0.851677371$	$y(0.5) = 0.883727921$	3.63 %

Example 3

Repeat the previous [example \(Example 2\)](#) only this time give the approximations at $t=1$, $t=2$, $t=3$, $t=4$, and $t=5$. Use $h=0.1$, $h=0.05$, $h=0.01$, $h=0.005$, and $h=0.001$ for the approximations.

Approximations						
Time	Exact	$h = 0.1$	$h = 0.05$	$h = 0.01$	$h = 0.005$	$h = 0.001$
$t = 1$	0.9414902	0.9313244	0.9364698	0.9404994	0.9409957	0.9413914
$t = 2$	0.9910099	0.9913681	0.9911126	0.9910193	0.9910139	0.9910106
$t = 3$	0.9987637	0.9990501	0.9988982	0.9987890	0.9987763	0.9987662
$t = 4$	0.9998323	0.9998976	0.9998657	0.9998390	0.9998357	0.9998330
$t = 5$	0.9999773	0.9999890	0.9999837	0.9999786	0.9999780	0.9999774

Example 3 (cont.)

Repeat the previous example only this time give the approximations at $t=1$, $t=2$, $t=3$, $t=4$, and $t=5$. Use $h=0.1$, $h=0.05$, $h=0.01$, $h=0.005$, and $h=0.001$ for the approximations.

Percentage Errors					
Time	$h = 0.1$	$h = 0.05$	$h = 0.01$	$h = 0.005$	$h = 0.001$
$t = 1$	1.08 %	0.53 %	0.105 %	0.053 %	0.0105 %
$t = 2$	0.036 %	0.010 %	0.00094 %	0.00041 %	0.0000703 %
$t = 3$	0.029 %	0.013 %	0.0025 %	0.0013 %	0.00025 %
$t = 4$	0.0065 %	0.0033 %	0.00067 %	0.00034 %	0.000067 %
$t = 5$	0.0012 %	0.00064 %	0.00013 %	0.000068 %	0.000014 %

We can see from these tables that decreasing h does in fact improve the accuracy of the approximation as we expected.

Example 4

Consider the **Initial Value Problem (IVP)**

$$y' - y = -\frac{1}{2}e^{\frac{t}{2}}\sin(5t) + 5e^{\frac{t}{2}}\cos(5t) \quad y(0) = 0$$

Use Euler's Method to find the approximation to the solution at $t=1, t=2, t=3, t=4$, and $t=5$.
Use $h=0.1, h=0.05, h=0.01, h=0.005$, and $h=0.001$ for the approximations.

$y(t) = e^{\frac{t}{2}} \sin(5t)$	Approximations						
	Time	Exact	$h = 0.1$	$h = 0.05$	$h = 0.01$	$h = 0.005$	$h = 0.001$
	$t = 1$	-1.58100	-0.97167	-1.26512	-1.51580	-1.54826	-1.57443
	$t = 2$	-1.47880	0.65270	-0.34327	-1.23907	-1.35810	-1.45453
	$t = 3$	2.91439	7.30209	5.34682	3.44488	3.18259	2.96851
	$t = 4$	6.74580	15.56128	11.84839	7.89808	7.33093	6.86429
	$t = 5$	-1.61237	21.95465	12.24018	1.56056	0.0018864	-1.28498

Example 4 (cont.)

Consider the **Initial Value Problem (IVP)**

$$y' - y = -\frac{1}{2}e^{\frac{t}{2}} \sin(5t) + 5e^{\frac{t}{2}} \cos(5t) \quad y(0) = 0$$

Use Euler's Method to find the approximation to the solution at $t=1$, $t=2$, $t=3$, $t=4$, and $t=5$.
Use $h=0.1$, $h=0.05$, $h=0.01$, $h=0.005$, and $h=0.001$ for the approximations.

$$y(t) = e^{\frac{t}{2}} \sin(5t)$$

Percentage Errors					
Time	$h = 0.1$	$h = 0.05$	$h = 0.01$	$h = 0.005$	$h = 0.001$
$t = 1$	38.54 %	19.98 %	4.12 %	2.07 %	0.42 %
$t = 2$	144.14 %	76.79 %	16.21 %	8.16 %	1.64 %
$t = 3$	150.55 %	83.46 %	18.20 %	9.20 %	1.86 %
$t = 4$	130.68 %	75.64 %	17.08 %	8.67 %	1.76 %
$t = 5$	1461.63 %	859.14 %	196.79 %	100.12 %	20.30 %

Heun's Method

- A fundamental source of error in Euler's method is that the derivative at the beginning of the interval is assumed to apply across the entire interval.
- Two simple modifications are available to help circumvent this shortcoming.
- One method to improve the estimate of the slope involves the determination of two derivatives for the interval—one at the initial point and another at the end point
- The two derivatives are then averaged to obtain an improved estimate of the slope for the entire interval. This approach, called *Heun's method*,

Heun's Method

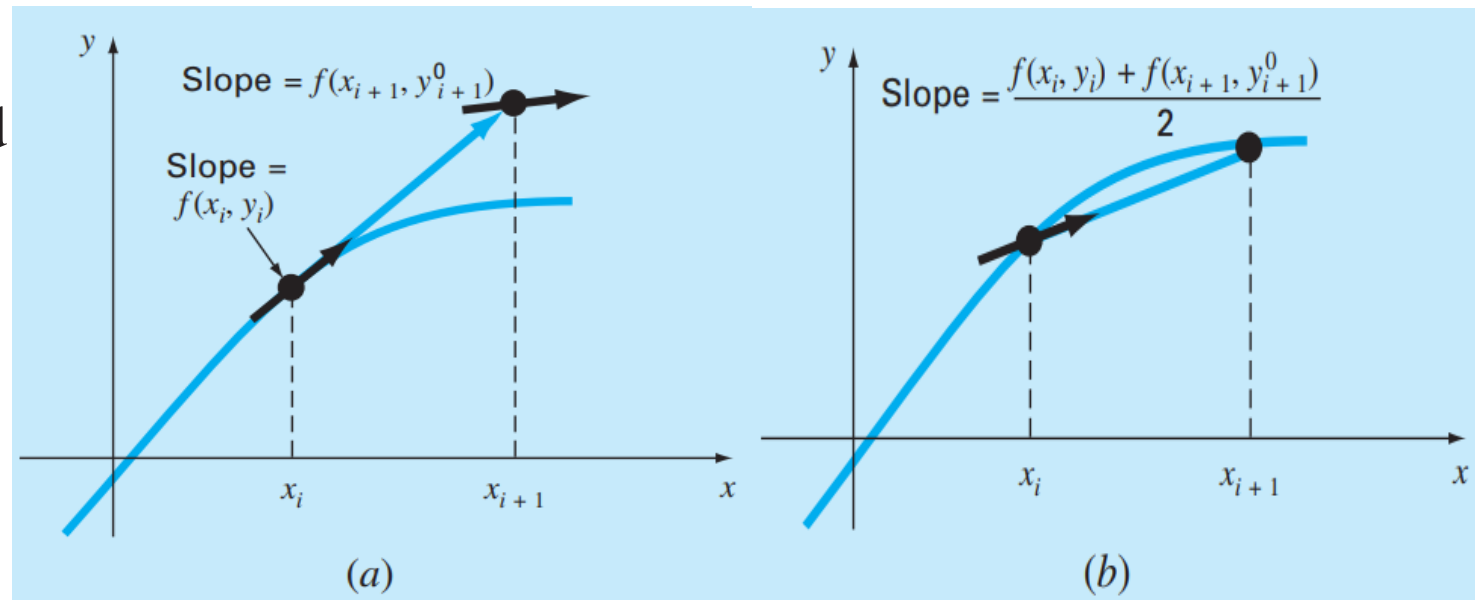
- In Euler's method, the slope at the beginning of an interval

$$y'_i = f(x_i, y_i)$$

is used to extrapolate linearly to y_{i+1} :

$$y_{i+1}^0 = y_i + f(x_i, y_i)h$$

For the standard Euler method we would stop at this point. However, in Heun's method the calculated y_{i+1}^0 is not the final answer, but **an intermediate prediction**. This is why we have distinguished it with a superscript 0. This is called a **predictor equation**.



Heun's Method

Past Equation provides an estimate of y_{i+1} that allows the calculation of an estimated slope at the end of the interval.

$$y'_{i+1} = f(x_{i+1}, y_{i+1}^0)$$

Thus, the two slopes can be combined to obtain an average slope for the interval:

$$\bar{y}' = \frac{y'_i + y'_{i+1}}{2} = \frac{f(x_i, y_i) + f(x_{i+1}, y_{i+1}^0)}{2}$$

This average slope is then used to extrapolate linearly from y_i to y_{i+1} using Euler's method:

$$y_{i+1} = y_i + \frac{f(x_i, y_i) + f(x_{i+1}, y_{i+1}^0)}{2}h$$

which is called a *corrector equation*. The **Heun method** is a *predictor-corrector approach*. It can be expressed concisely as

$$\begin{aligned} y_{i+1}^0 &= y_i + f(x_i, y_i)h \\ y_{i+1} &= y_i + \frac{f(x_i, y_i) + f(x_{i+1}, y_{i+1}^0)}{2}h \end{aligned}$$

Example 5

Problem Statement. Use Heun's method to integrate $y' = 4e^{0.8x} - 0.5y$ from $x = 0$ to $x = 4$ with a step size of 1. The initial condition at $x = 0$ is $y = 2$.

By using calculus, we can determine the following analytical solution.

$$y = \frac{4}{1.3}(e^{0.8x} - e^{-0.5x}) + 2e^{-0.5x}$$

Solution. First, the slope at (x_0, y_0) is calculated as

$$y'_0 = 4e^0 - 0.5(2) = 3$$

The numerical solution is obtained by using the predictor to obtain an estimate of y at 1.0:

$$y_1^0 = 2 + 3(1) = 5$$

Now, to improve the estimate for y_{i+1} , we use the value y_1^0 to predict the slope at the end of the interval

$$y'_1 = f(x_1, y_1^0) = 4e^{0.8(1)} - 0.5(5) = 6.402164$$

Example 5 (cont.)

which can be combined with the initial slope to yield an average slope over the interval from $x = 0$ to 1

$$y' = \frac{3 + 6.402164}{2} = 4.701082$$

This result can then be substituted into the corrector to give the prediction at $x = 1$

$$y_1 = 2 + 4.701082(1) = 6.701082$$

x	y_{true}	y_{Heun}	 ε_t (%)
0	2.0000000	2.0000000	0.00
1	6.1946314	6.7010819	8.18
2	14.8439219	16.3197819	9.94
3	33.6771718	37.1992489	10.46
4	75.3389626	83.3377674	10.62

Example 6

Let's use the same IVP as before:

$$y' + 2y = 2 - e^{-4t}, \quad y(0) = 1$$

$$y' = f(t, y) = 2 - e^{-4t} - 2y$$

Step size: $h = 0.1$

We'll find approximate values at $t = 0.1, 0.2, 0.3, 0.4, 0.5$

Initial Condition

$$t_0 = 0, \quad y_0 = 1$$

Step 1: $t = 0.1$

Predictor:

$$f(t_0, y_0) = 2 - e^0 - 2(1) = 2 - 1 - 2 = -1$$

$$\tilde{y}_1 = y_0 + h \cdot f(t_0, y_0) = 1 + 0.1 \cdot (-1) = 0.9$$

Corrector:

$$f(t_1, \tilde{y}_1) = 2 - e^{-0.4} - 2(0.9)$$

$$e^{-0.4} \approx 0.670320$$

$$f(t_1, \tilde{y}_1) = 2 - 0.670320 - 1.8 = -0.470320$$

$$y_1 = y_0 + \frac{h}{2} [f(t_0, y_0) + f(t_1, \tilde{y}_1)]$$

$$y_1 = 1 + \frac{0.1}{2} [-1 + (-0.470320)]$$

$$y_1 = 1 + 0.05 \cdot (-1.470320) = 1 - 0.073516 = 0.926484$$

Example 6 (cont.)

Step 2: $t = 0.2$

Predictor:

$$\begin{aligned}f(t_1, y_1) &= 2 - e^{-0.4} - 2(0.926484) \\&= 2 - 0.670320 - 1.852968 = -0.523288\end{aligned}$$

$$\tilde{y}_2 = y_1 + h \cdot f(t_1, y_1) = 0.926484 + 0.1 \cdot (-0.523288) = 0.874155$$

Corrector:

$$\begin{aligned}f(t_2, \tilde{y}_2) &= 2 - e^{-0.8} - 2(0.874155) \\e^{-0.8} &\approx 0.449329\end{aligned}$$

$$f(t_2, \tilde{y}_2) = 2 - 0.449329 - 1.748310 = -0.197639$$

$$y_2 = y_1 + \frac{h}{2} [f(t_1, y_1) + f(t_2, \tilde{y}_2)]$$

$$y_2 = 0.926484 + 0.05 [-0.523288 + (-0.197639)]$$

$$y_2 = 0.926484 + 0.05 \cdot (-0.720927) = 0.926484 - 0.036046 = 0.890438$$

Step 3: $t = 0.3$

Predictor:

$$\begin{aligned}f(t_2, y_2) &= 2 - e^{-0.8} - 2(0.890438) \\&= 2 - 0.449329 - 1.780876 = -0.230205\end{aligned}$$

$$\tilde{y}_3 = y_2 + h \cdot f(t_2, y_2) = 0.890438 + 0.1 \cdot (-0.230205) = 0.867417$$

Corrector:

$$\begin{aligned}f(t_3, \tilde{y}_3) &= 2 - e^{-1.2} - 2(0.867417) \\e^{-1.2} &\approx 0.301194\end{aligned}$$

$$f(t_3, \tilde{y}_3) = 2 - 0.301194 - 1.734834 = -0.036028$$

$$y_3 = y_2 + \frac{h}{2} [f(t_2, y_2) + f(t_3, \tilde{y}_3)]$$

$$y_3 = 0.890438 + 0.05 [-0.230205 + (-0.036028)]$$

Example 6 (cont.)

Step 2: $t = 0.2$

Predictor:

$$\begin{aligned}f(t_1, y_1) &= 2 - e^{-0.4} - 2(0.926484) \\&= 2 - 0.670320 - 1.852968 = -0.523288\end{aligned}$$

$$\tilde{y}_2 = y_1 + h \cdot f(t_1, y_1) = 0.926484 + 0.1 \cdot (-0.523288) = 0.874155$$

Corrector:

$$\begin{aligned}f(t_2, \tilde{y}_2) &= 2 - e^{-0.8} - 2(0.874155) \\e^{-0.8} &\approx 0.449329\end{aligned}$$

$$f(t_2, \tilde{y}_2) = 2 - 0.449329 - 1.748310 = -0.197639$$

$$y_2 = y_1 + \frac{h}{2} [f(t_1, y_1) + f(t_2, \tilde{y}_2)]$$

$$y_2 = 0.926484 + 0.05 [-0.523288 + (-0.197639)]$$

$$y_2 = 0.926484 + 0.05 \cdot (-0.720927) = 0.926484 - 0.036046 = 0.890438$$

Step 3: $t = 0.3$

Predictor:

$$\begin{aligned}f(t_2, y_2) &= 2 - e^{-0.8} - 2(0.890438) \\&= 2 - 0.449329 - 1.780876 = -0.230205\end{aligned}$$

$$\tilde{y}_3 = y_2 + h \cdot f(t_2, y_2) = 0.890438 + 0.1 \cdot (-0.230205) = 0.867417$$

Corrector:

$$\begin{aligned}f(t_3, \tilde{y}_3) &= 2 - e^{-1.2} - 2(0.867417) \\e^{-1.2} &\approx 0.301194\end{aligned}$$

$$f(t_3, \tilde{y}_3) = 2 - 0.301194 - 1.734834 = -0.036028$$

$$y_3 = y_2 + \frac{h}{2} [f(t_2, y_2) + f(t_3, \tilde{y}_3)]$$

$$y_3 = 0.890438 + 0.05 [-0.230205 + (-0.036028)]$$

Example 6 (cont.)

Step 4: $t = 0.4$

Predictor:

$$\begin{aligned}f(t_3, y_3) &= 2 - e^{-1.2} - 2(0.877126) \\&= 2 - 0.301194 - 1.754252 = -0.055446\end{aligned}$$

$$\tilde{y}_4 = y_3 + h \cdot f(t_3, y_3) = 0.877126 + 0.1 \cdot (-0.055446) = 0.871581$$

Corrector:

$$\begin{aligned}f(t_4, \tilde{y}_4) &= 2 - e^{-1.6} - 2(0.871581) \\e^{-1.6} &\approx 0.201897 \\f(t_4, \tilde{y}_4) &= 2 - 0.201897 - 1.743162 = 0.054941\end{aligned}$$

$$y_4 = y_3 + \frac{h}{2} [f(t_3, y_3) + f(t_4, \tilde{y}_4)]$$

$$y_4 = 0.877126 + 0.05 [-0.055446 + 0.054941]$$

$$y_4 = 0.877126 + 0.05 \cdot (-0.000505) = 0.877126 - 0.000025 = 0.877101$$

Step 5: $t = 0.5$

Predictor:

$$\begin{aligned}f(t_4, y_4) &= 2 - e^{-1.6} - 2(0.877101) \\&= 2 - 0.201897 - 1.754202 = 0.043901\end{aligned}$$

$$\tilde{y}_5 = y_4 + h \cdot f(t_4, y_4) = 0.877101 + 0.1 \cdot 0.043901 = 0.881491$$

Corrector:

$$\begin{aligned}f(t_5, \tilde{y}_5) &= 2 - e^{-2.0} - 2(0.881491) \\e^{-2.0} &\approx 0.135335 \\f(t_5, \tilde{y}_5) &= 2 - 0.135335 - 1.762982 = 0.101683\end{aligned}$$

$$y_5 = y_4 + \frac{h}{2} [f(t_4, y_4) + f(t_5, \tilde{y}_5)]$$

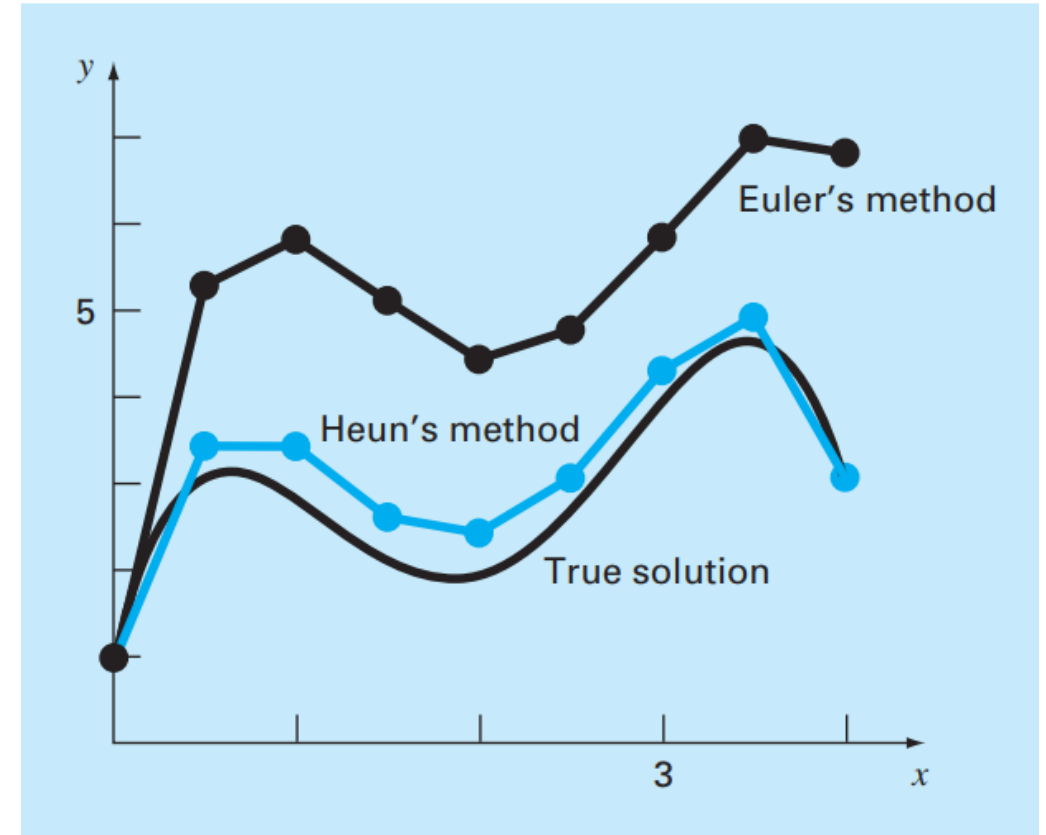
$$y_5 = 0.877101 + 0.05 [0.043901 + 0.101683]$$

$$y_5 = 0.877101 + 0.05 \cdot 0.145584 = 0.877101 + 0.007279 = 0.884380$$

Example 6 (cont.)

Summary of Results

t	Heun's Method	Exact Solution	Error
0.0	1.000000	1.000000	0.000000
0.1	0.926484	0.925795	0.000689
0.2	0.890438	0.889505	0.000933
0.3	0.877126	0.876191	0.000935
0.4	0.877101	0.876284	0.000817
0.5	0.884380	0.883728	0.000652



Comparison of the true solution with a numerical solution using Euler's and Heun's methods for the integral of $y' = -2x^3 + 12x^2 - 20x + 8.5$.

The Midpoint (or Improved Polygon) Method

- Another simple modification of Euler's method is called the *midpoint method* (or the *improved polygon* or the *modified Euler*). This technique uses Euler's method to predict a value of y at the midpoint of the interval.

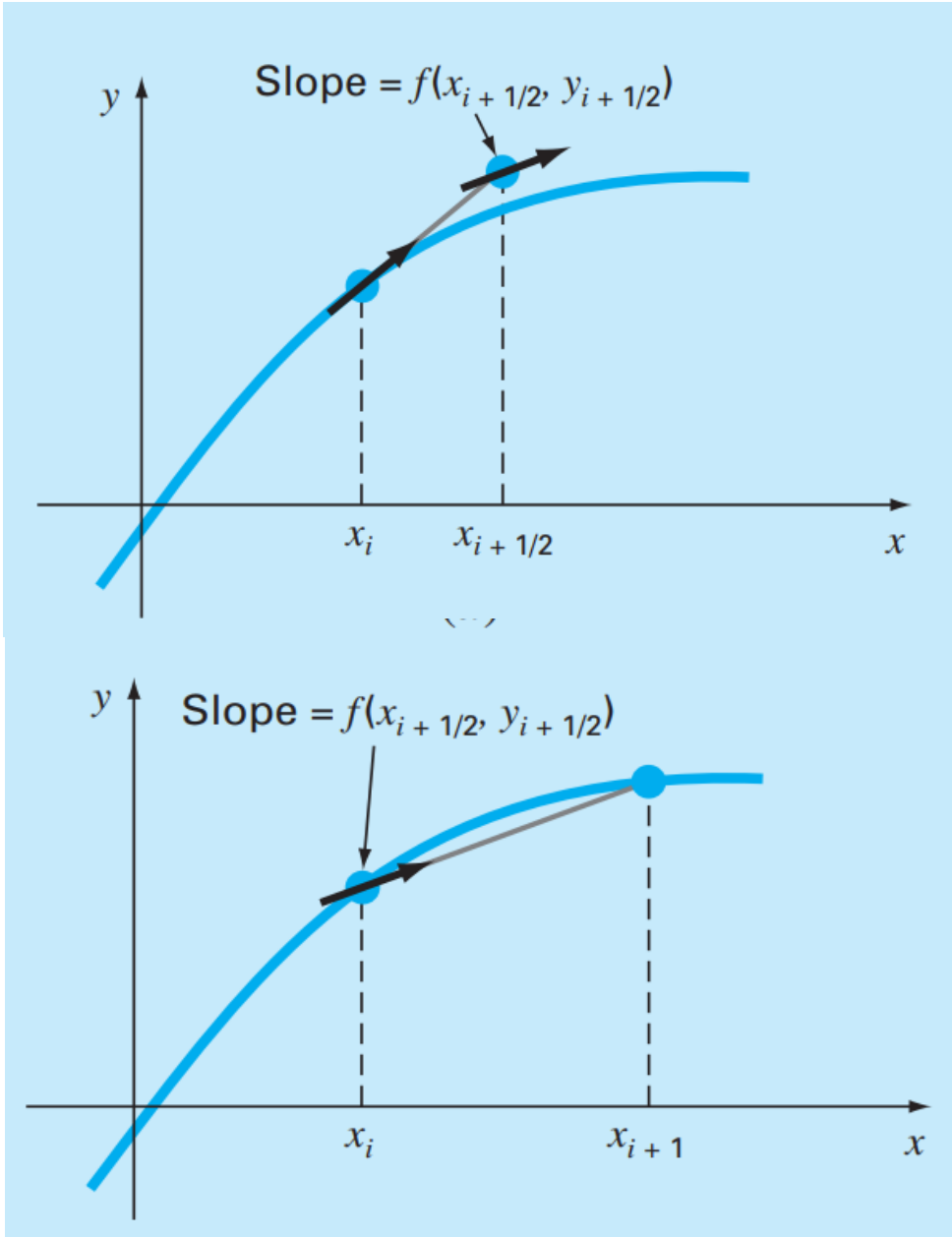
$$y_{i+1/2} = y_i + f(x_i, y_i) \frac{h}{2}$$

- Then, this predicted value is used to calculate a slope at the midpoint.

$$y'_{i+1/2} = f(x_{i+1/2}, y_{i+1/2})$$

$$y_{i+1} = y_i + f(x_{i+1/2}, y_{i+1/2})h$$

The Midpoint (or Improved Polygon) Method



Midpoint Method Algorithm

For the ODE $y' = f(t, y)$:

1. Compute slope at current point:

$$k_1 = f(t_n, y_n)$$

2. Compute midpoint prediction:

$$t_{mid} = t_n + \frac{h}{2}$$

$$y_{mid} = y_n + \frac{h}{2} \cdot k_1$$

3. Compute slope at midpoint:

$$k_2 = f(t_{mid}, y_{mid})$$

4. Take full step using midpoint slope:

$$y_{n+1} = y_n + h \cdot k_2$$

$$t_{n+1} = t_n + h$$

Example 7

Let's solve the IVP:

$$y' = t - y + 1, \quad y(0) = 1$$

We'll use step size $h = 0.2$ and find approximate values at $t = 0.2, 0.4, 0.6, 0.8, 1.0$.

Exact Solution (for comparison):

This is a linear ODE with solution:

$$y(t) = t + e^{-t}$$

Initial Condition

$$t_0 = 0, \quad y_0 = 1$$

Step 1: $t = 0.2$

Slope at current point:

$$k_1 = f(t_0, y_0) = 0 - 1 + 1 = 0$$

Midpoint prediction:

$$t_{mid} = 0 + \frac{0.2}{2} = 0.1$$

$$y_{mid} = 1 + \frac{0.2}{2} \cdot 0 = 1 + 0.1 \cdot 0 = 1$$

Slope at midpoint:

$$k_2 = f(t_{mid}, y_{mid}) = 0.1 - 1 + 1 = 0.1$$

Full step:

$$y_1 = y_0 + h \cdot k_2 = 1 + 0.2 \cdot 0.1 = 1.02$$

$$t_1 = 0.2$$

Exact value: $y(0.2) = 0.2 + e^{-0.2} = 0.2 + 0.818731 = 1.018731$

Example 7 (Cont.)

Step 2: $t = 0.4$

Slope at current point:

$$k_1 = f(t_1, y_1) = 0.2 - 1.02 + 1 = 0.18$$

Midpoint prediction:

$$t_{mid} = 0.2 + \frac{0.2}{2} = 0.3$$

$$y_{mid} = 1.02 + \frac{0.2}{2} \cdot 0.18 = 1.02 + 0.1 \cdot 0.18 = 1.038$$

Slope at midpoint:

$$k_2 = f(t_{mid}, y_{mid}) = 0.3 - 1.038 + 1 = 0.262$$

Full step:

$$y_2 = y_1 + h \cdot k_2 = 1.02 + 0.2 \cdot 0.262 = 1.0724$$

$$t_2 = 0.4$$

Exact value: $y(0.4) = 0.4 + e^{-0.4} = 0.4 + 0.670320 = 1.070320$

Step 3: $t = 0.6$

Slope at current point:

$$k_1 = f(t_2, y_2) = 0.4 - 1.0724 + 1 = 0.3276$$

Midpoint prediction:

$$t_{mid} = 0.4 + \frac{0.2}{2} = 0.5$$

$$y_{mid} = 1.0724 + \frac{0.2}{2} \cdot 0.3276 = 1.0724 + 0.1 \cdot 0.3276 = 1.10516$$

Slope at midpoint:

$$k_2 = f(t_{mid}, y_{mid}) = 0.5 - 1.10516 + 1 = 0.39484$$

Full step:

$$y_3 = y_2 + h \cdot k_2 = 1.0724 + 0.2 \cdot 0.39484 = 1.151368$$

$$t_3 = 0.6$$

Exact value: $y(0.6) = 0.6 + e^{-0.6} = 0.6 + 0.548812 = 1.148812$

Example 7 (Cont.)

Step 4: $t = 0.8$

Slope at current point:

$$k_1 = f(t_3, y_3) = 0.6 - 1.151368 + 1 = 0.448632$$

Midpoint prediction:

$$t_{mid} = 0.6 + \frac{0.2}{2} = 0.7$$

$$y_{mid} = 1.151368 + \frac{0.2}{2} \cdot 0.448632 = 1.151368 + 0.1 \cdot 0.448632 = 1.196231$$

Slope at midpoint:

$$k_2 = f(t_{mid}, y_{mid}) = 0.7 - 1.196231 + 1 = 0.503769$$

Full step:

$$y_4 = y_3 + h \cdot k_2 = 1.151368 + 0.2 \cdot 0.503769 = 1.252122$$

$$t_4 = 0.8$$

Exact value: $y(0.8) = 0.8 + e^{-0.8} = 0.8 + 0.449329 = 1.249329$

Step 5: $t = 1.0$

Slope at current point:

$$k_1 = f(t_4, y_4) = 0.8 - 1.252122 + 1 = 0.547878$$

Midpoint prediction:

$$t_{mid} = 0.8 + \frac{0.2}{2} = 0.9$$

$$y_{mid} = 1.252122 + \frac{0.2}{2} \cdot 0.547878 = 1.252122 + 0.1 \cdot 0.547878 = 1.306910$$

Slope at midpoint:

$$k_2 = f(t_{mid}, y_{mid}) = 0.9 - 1.306910 + 1 = 0.593090$$

Full step:

$$y_5 = y_4 + h \cdot k_2 = 1.252122 + 0.2 \cdot 0.593090 = 1.370740$$

$$t_5 = 1.0$$

Exact value: $y(1.0) = 1.0 + e^{-1.0} = 1.0 + 0.367879 = 1.367879$

Example 7 (Cont.)

Summary of Results

t	Midpoint Method	Exact Solution	Error
0.0	1.000000	1.000000	0.000000
0.2	1.020000	1.018731	0.001269
0.4	1.072400	1.070320	0.002080
0.6	1.151368	1.148812	0.002556
0.8	1.252122	1.249329	0.002793
1.0	1.370740	1.367879	0.002861